

DP-ES: Differentially Private Evolution Strategies for Prompt Optimization

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Abstract

Token-level differentially private (DP) prompt optimization methods such as DP-OPT exhibit systematic failure modes under tight privacy budgets: on GSM8K with $\epsilon = 1.0$, a 30-run analysis reveals a **63% failure rate**, driven by *template damage*—the search trajectory deletes structural placeholders such as {question}—and noise-driven *suboptimal selection*. We diagnose these as structural consequences of greedy token-by-token construction over privately aggregated counts. We then propose **DP-ES** (Differentially Private Evolution Strategies), a structurally cleaner alternative that maintains a population of full prompts, mutates them via LLM calls that never access the private dataset, and spends privacy only on sampled-Gaussian evaluation; deterministic or Gumbel-smoothed selection is post-processing. Under a conservative ($\epsilon \leq 1.0$, $\delta = 10^{-5}$) guarantee, DP-ES achieves **88.1% on GSM8K** (+38.6 pp over DP-OPT, $\sim 9\times$ lower std), **99.7% on MedQA**, **73.5% on BANKING77**, and **86.8% on Alpaca**, with **0/30 operational failures on GSM8K**, $2.5\times$ faster wall-clock runtime, and $3.3\times$ fewer logged private-data call groups than DP-OPT. Selection and population ablations, implementation-level noise checks, and a 200-profile exact-match memorization stress test complement the formal guarantee. **Scope.** Our experiments establish optimization robustness under DP noise, especially where prompt structure is critical; end-to-end validation on genuinely sensitive, non-saturated deployment data remains future work. Code: github.com/StephCpa/dp-es.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities through in-context learning (Brown et al., 2020; Ouyang et al., 2022), yet their performance heavily relies on carefully crafted prompts (Liu et al., 2023). Automatic prompt optimization (Yuksekgonul et al., 2024;

Yang et al., 2024b) can bridge this gap, but when the optimization data are sensitive (e.g., patient records, financial logs), the optimizer can memorize and expose private information, as evidenced by recent membership inference and extraction attacks (Wang et al., 2023; Shanmugarasa et al., 2025; Fu et al., 2023). For instance, a naive optimizer might generate a prompt like “Remember how patient #12847 had atrial fibrillation...”, directly violating privacy. Any optimizer intended for such settings must therefore satisfy formal **Differential Privacy (DP)** guarantees (Dwork et al., 2006)—and, crucially, the optimization mechanism itself must remain robust under the noise that DP introduces, which is the question we focus on in this work.

The Challenge: Task Complexity and Privacy Budgets. Applying DP to prompt optimization is non-trivial due to the discrete nature of text. Recent work like DP-OPT (Hong et al., 2024) uses histogram-based token generation with the exponential mechanism. Other approaches apply DP to in-context learning via private few-shot generation or nearest neighbor retrieval (Tang et al., 2024; Koskela et al., 2025), but they rely on additional retrieval or embedding machinery. While DP-OPT is effective on simple tasks, we identify critical limitations on complex reasoning: our analysis of 30 DP-OPT runs reveals a **63% failure rate** on GSM8K (Appendix C.2, Table 8), with systematic issues including template damage (destruction or partial loss) and suboptimal selection. This fragility stems from token-level myopia and non-backtracking search, which become catastrophic under tight privacy constraints.

Private Evolution Perspective. Private Evolution (PE) is a training-free DP paradigm for data synthesis via APIs (Lin et al., 2024; Xie et al., 2024; Lin et al., 2025). Its core advantage is structural: only low-sensitivity evaluation touches private data, while exploration/generation and se-

lection over privatized scores are post-processing. This yields a decoupling between exploration and privacy expenditure—increasing the number or strength of mutations improves search without consuming additional privacy budget. DP-ES is a prompt-optimization instantiation of this PE principle.

Our Contribution: DP-ES. We frame this work as a *diagnostic and methodological* contribution for DP prompt optimization: we systematically characterize the failure modes of token-level DP construction under tight privacy budgets and propose **Differentially Private Evolution Strategies (DP-ES)**, a structurally cleaner alternative. Unlike methods that build prompts token by token from privately aggregated counts, DP-ES maintains a population of full prompts that evolve over time. The key design choice is to decouple exploration from privacy consumption: using LLMs to generate semantic mutations consumes **zero privacy budget** (by the post-processing property), allowing the scarce privacy budget to be allocated strictly to sampled-Gaussian evaluation.

Our contributions are:

1. **Failure-mode diagnosis:** A 30-run analysis of DP-OPT on GSM8K identifying *template damage* (placeholder loss, 26% of failures) and *noise-driven suboptimal selection* (74%) as structural failure modes of token-level DP prompt construction, supported by concrete trajectory evidence (Section 5).
2. **Method:** A structurally cleaner evolution-based DP prompt optimizer with privacy-free mutations, sampled-Gaussian scoring under RDP accounting, and deterministic or Gumbel-smoothed post-processing selection.
3. **Empirical robustness:** 0/30 operational failures on GSM8K (vs. 19/30 for DP-OPT), with $\sim 9\times$ lower standard deviation, plus strong or competitive results on MedQA, BANKING77, and Alpaca under a conservative ($\epsilon \leq 1, \delta = 10^{-5}$) guarantee.
4. **Efficiency:** $2.5\times$ faster wall-clock runtime and $3.3\times$ fewer logged private-data call groups than DP-OPT on GSM8K, with an explicit generation-vs-evaluation cost decomposition.
5. **Privacy evidence:** A reproducible RDP accounting calculation and noise-distribution checks, plus a 200-profile synthetic exact-match stress test (0 PII strings found for DP-ES vs. 40 for an adversarial non-DP control).

Scope of Empirical Evidence. Our experiments establish DP-ES as a *structurally more robust DP prompt optimizer* than token-level alternatives, especially on tasks where prompt structure is critical. They do not by themselves validate deployment on genuinely sensitive, non-saturated datasets. The formal (ϵ, δ) -DP guarantee bounds record-level leakage independently of the benchmark, while the implementation checks and exact-match stress test provide complementary diagnostics. We make these scope boundaries explicit in the Limitations section.

2 Related Work

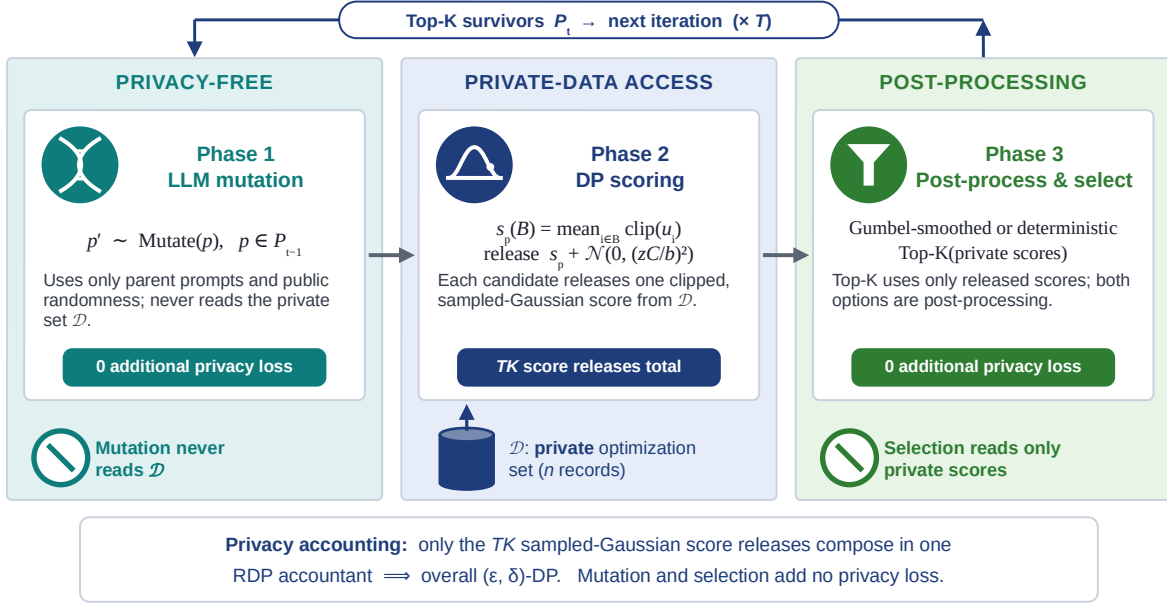
Prompt Optimization. Gradient-free methods such as APE (Zhou et al., 2023) and OPRO (Yang et al., 2024b) search over discrete prompt candidates using LLM-generated variations, while gradient-based approaches like TextGrad (Yuksekgonul et al., 2024) use LLM feedback as differentiable signals. Both families optimize prompts effectively but assume unrestricted access to training data with no privacy guarantees (Edemacu and Wu, 2025), making them unsuitable for sensitive domains such as healthcare, finance, and education where data protection is legally mandated.

Differential Privacy in Machine Learning. DP-SGD (Abadi et al., 2016) and PATE (Papernot et al., 2018) provide formal (ϵ, δ) -DP guarantees for continuous model parameters during training. DP-LoRA (Liu et al., 2024b; Frery et al., 2025) extends this to parameter-efficient fine-tuning of LLMs, achieving strong performance with reduced memory overhead. However, all these methods require access to model weights, precluding their use with black-box API-only LLMs—the dominant deployment mode for state-of-the-art models such as GPT-4 and DeepSeek.

DP Prompt Optimization. For discrete prompts over black-box LLMs, DP-OPT (Hong et al., 2024) uses histogram-based token generation with the exponential mechanism, constructing prompts token by token from privately aggregated counts. While effective on classification and factual QA, this greedy token-level approach suffers from template damage and suboptimal selection on complex reasoning tasks (63% failure rate on GSM8K; Section 5), as noise accumulates across sequential token decisions without the ability to backtrack.

Other Approaches. Other directions include DP few-shot demonstration generation for in-context

(a) One DP-ES iteration: privacy-free exploration, private scoring



(b) DP-ES vs. DP-OPT: mechanism and observed stability

Aspect	DP-ES (ours)	DP-OPT
Search unit	Population of complete prompts	Greedy token-by-token construction
Private-data access	Sampled utility scoring only	Private histogram queries during construction
Selection / recovery	Top-K on privatized scores; alternatives persist	Private token releases; no token-level backtracking
GSM8K (30 runs)	88.1 $\pm$ 3.2%	49.5 $\pm$ 28.5%

Figure 1: **DP-ES workflow and mechanism-level contrast with DP-OPT.** Only sampled-Gaussian scoring accesses the private dataset. LLM mutation and deterministic or Gumbel-smoothed top- K selection operate without further private-data access and add no privacy loss. Mechanism rows summarize canonical DP-OPT; the bottom panel reports our black-box adaptation’s GSM8K mean and standard deviation across 30 runs without imposing a post-hoc failure threshold.

learning (Tang et al., 2024), which privately selects exemplars but cannot optimize the instruction prompt itself; private retrieval augmentation (Koskela et al., 2025), which protects retrieval queries but inherits the limitations of fixed prompt templates; and PromptDPSGD (Duan et al., 2023), which applies DP-SGD to continuous soft prompt embeddings but requires white-box model access and struggles on smaller models (Appendix C.3).

Private Evolution. Private Evolution (PE) (Lin et al., 2024; Xie et al., 2024; Lin et al., 2025) is a training-free DP framework originally developed for data synthesis via APIs. Its key structural advantage is that only a low-sensitivity evaluation step touches private data, while API-based generation is pure post-processing. This creates a favorable decoupling: exploration can be arbitrarily rich without consuming privacy budget. DP-ES adapts this principle from data synthesis

to prompt optimization, inheriting the favorable privacy-exploration trade-off while introducing population-based search over the discrete prompt space.

Evolution Strategies and DP. Evolution strategies (ES) (Rechenberg, 1973) are a family of derivative-free optimization algorithms that maintain and evolve populations of candidate solutions. ES have driven advances in neural architecture search (Real et al., 2019) and reinforcement learning (Jaderberg et al., 2017), demonstrating strong performance in high-dimensional, noisy search spaces. While prior work has combined DP with optimization algorithms such as ERM (Chaudhuri et al., 2011), DP-ES is the first to leverage evolutionary search for DP prompt optimization in the discrete text domain. The population-based design provides natural robustness to noise-induced regressions—a critical advantage under tight privacy budgets

Method	Priv.	Disc.	Mechanism
TextGrad	×	✓	LLM gradients
OPRO	×	✓	LLM scoring
DP-SGD	✓	×	Gaussian noise
DP-OPT	✓	✓	Histogram + Exp
DP-ES	✓	✓	Evol. + Gauss

Table 1: Comparison of prompt optimization methods. DP-ES uniquely combines evolution strategies with DP for discrete prompts.

$\mathcal{D}, n$	private data, size	K, T	population, rounds
b, q	batch size, $q = b/n$	z	Gaussian noise multiplier
$\tilde{s}_p$	privatized score	β	Gumbel smoothing scale

Table 2: Notation used in DP-ES.

where evaluation noise is substantial.

Privacy Auditing. Empirical privacy attacks and audits can reveal implementation errors but cannot certify DP from a finite number of trials (Nasr et al., 2021; Jagielski et al., 2020). We therefore base our guarantee on mechanism-level RDP accounting and use noise checks and an exact-match memorization stress test only as complementary diagnostics (Section 6).

Positioning of DP-ES. Table 1 summarizes the landscape. DP-ES uniquely combines evolutionary exploration with sampled-Gaussian scoring and post-processing selection, filling the gap for a robust, black-box-compatible DP prompt optimization method that scales across diverse task complexities—from structured classification to multi-step reasoning.

3 Methodology: DP-ES

Problem Formulation. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a private dataset where x_i represents task inputs and y_i represents desired outputs (e.g., question-answer pairs). Let $\mathcal{P}$ denote the space of natural language prompts. Our goal is to find a prompt $p^* \in \mathcal{P}$ that maximizes task performance:

$$p^* = \arg \max_{p \in \mathcal{P}} f(p, \mathcal{D}) \quad (1)$$

where $f : \mathcal{P} \times \mathcal{D} \rightarrow \mathbb{R}$ measures prompt quality (e.g., accuracy on $\mathcal{D}$).

Privacy requirement: The optimization algorithm must satisfy (ϵ, δ) -differential privacy with respect to $\mathcal{D}$, ensuring that any single record (x_i, y_i) has bounded influence on the output prompt.

Definition 1 ((ϵ, δ) -Differential Privacy). A randomized algorithm $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{O}$ satisfies (ϵ, δ) -DP if for all neighboring datasets $\mathcal{D} \sim \mathcal{D}'$ (differing in one record) and all measurable sets $S \subseteq \text{Range}(\mathcal{M})$:

Algorithm 1 DP-ES: Differentially Private Evolution Strategies

Require: Private dataset $\mathcal{D}$, privacy budget $(\epsilon_{\text{total}}, \delta)$, initial prompt p_0 , population size K , iterations T
Ensure: Optimized prompt p^* with $(\epsilon_{\text{total}}, \delta)$ -DP
1: Initialize population $\mathcal{P}_0 \leftarrow \{p_0\}$
2: Calibrate z for TK sampled-Gaussian scores {RDP accountant}
3: **for** $t = 1$ to T **do**
4: *// Phase 1: Privacy-free mutation*
5: **for** each $p \in \mathcal{P}_{t-1}$ **do**
6: Generate K children: $\{p'_1, \dots, p'_K\} \sim \text{LLM-Mutate}(p)$
7: **end for**
8: $\mathcal{P}_t \leftarrow \mathcal{P}_{t-1} \cup \{\text{all children}\}$
9: *// Phase 2: Sampled-Gaussian scoring (RDP-accounted)*
10: **for** each $p \in \mathcal{P}_t$ **do**
11: Sample $B \subset \mathcal{D}$, $|B| = b$; $s_p = \frac{1}{b} \sum_{i \in B} \mathbb{I}[\text{LLM}(p, x_i) = y_i]$
12: Clip: $s'_p = \text{clip}(s_p, [0, C_{\text{clip}}])$ $\{C_{\text{clip}} = 1.0\}$
13: Add noise: $\tilde{s}_p = s'_p + \mathcal{N}(0, \sigma^2)$ $\{\sigma = zC_{\text{clip}}/b\}$
14: **end for**
15: *// Phase 3: Selection over privatized scores*
16: **Option A** (default): $\hat{s}_p \leftarrow \tilde{s}_p + \text{Gumbel}(0, \beta)$ {randomized post-processing}
17: **Option B:** $\hat{s}_p \leftarrow \tilde{s}_p$ {Deterministic top- k ; pure post-processing}
18: Select top- K : $\mathcal{P}_{t+1} \leftarrow \text{top-}K(\mathcal{P}_t, \{\hat{s}_p\})$
19: **end for**
20: **return** $p^* \leftarrow \arg \max_{p \in \mathcal{P}_T} \hat{s}_p$

$$\mathbb{P}[\mathcal{M}(\mathcal{D}) \in S] \leq e^\epsilon \cdot \mathbb{P}[\mathcal{M}(\mathcal{D}') \in S] + \delta \quad (2)$$

DP-ES Algorithm Overview. DP-ES (Algorithm 1) employs an evolution strategy with three key components:

1. **Mutation:** Generate prompt variations using an LLM (privacy-free)
2. **DP Evaluation:** Score prompts on $\mathcal{D}$ with calibrated noise
3. **Selection:** Select top prompts from privatized scores (Gumbel-smoothed or deterministic)

Privacy-Free Mutations. Key Innovation: Unlike gradient-based methods that require DP protection for gradient computation, DP-ES sends only a public or previously privatized parent prompt to the mutation LLM; neither records from $\mathcal{D}$ nor record-derived feedback are included. Mutation is therefore post-processing and consumes zero additional privacy budget.

Proposition 1 (Mutation Privacy-Freeness). Let $\text{Mutate} : \mathcal{P} \rightarrow \mathcal{P}$ be a function that generates a new prompt from a parent prompt using an LLM, without accessing the private dataset $\mathcal{D}$. Then Mutate consumes zero privacy budget.

Proof. See Appendix D.

Proposition 2 (Decoupled Exploration). If mutation does not access $\mathcal{D}$, then the privacy loss

of DP-ES depends only on the sampled-Gaussian score releases and is independent of the number or strength of mutations per iteration. **Proof.** See Appendix D.

Mutation strategies: We implement 3 modes (EXPLORE, EXPLOIT, BALANCED); details and prompt templates are provided in Appendix A.

DP Evaluation via Gaussian Mechanism. After mutation, each candidate is evaluated on a uniformly sampled batch $B \subset \mathcal{D}$ of size b . Its score is the mean of clipped per-record utilities:

$$s_p(B) = \frac{1}{b} \sum_{i \in B} \mathbb{I}[\text{LLM}(p, x_i) = y_i] \in [0, 1]. \quad (3)$$

Clipping to $[0, C_{\text{clip}}]$ gives batch sensitivity $\Delta_B = C_{\text{clip}}/b$. We release $\tilde{s}_p = s_p + \mathcal{N}(0, (z\Delta_B)^2)$ and account for uniform sampling without replacement at rate $q = b/n$ using Rényi DP (RDP), composed over all TK scoring operations and converted to (ϵ, δ) -DP (Wang et al., 2019). In the reported optimization runs, $n=200$, $b=10$, $z=10$, $K=6$, and $T=3$; full-set variants use the corresponding $b=n$ sensitivity. AutoDP gives $\epsilon \leq 0.71$ at $\delta = 10^{-5}$, so all results conservatively satisfy the stated $\epsilon \leq 1$ guarantee. Full accounting details are in Appendix B.2.

Selection, Composition, End-to-End DP. Once the scores are privatized, both deterministic top- k (**Option B**) and our default Gumbel-smoothed top- k (**Option A**) are randomized post-processing (Dwork and Roth, 2014); neither accesses $\mathcal{D}$ or incurs additional privacy loss. The Gumbel perturbation is used only to smooth noise-induced rank inversions, not as a second privacy mechanism. Composing the TK sampled-Gaussian score releases with the RDP accountant therefore establishes end-to-end (ϵ, δ) -DP for Algorithm 1 (Theorem 3; Appendix D).

4 Experimental Setup

We evaluate DP-ES on four tasks spanning diverse task families: **GSM8K** (Cobbe et al., 2021) (multi-step math reasoning; 200-example optimization set and 1,319-example test set), **MedQA** (Jin et al., 2021) (medical multiple-choice QA, 200-question subset due to API cost constraints; DP-ES already achieves near-ceiling performance at 99.7% on this subset, limiting further discriminative power), **BANKING77** (Casanueva et al., 2020) (77-class intent classification, 200 examples), and **Alpaca** (Taori et al., 2023) (open-ended instruction-following, 200 examples scored by an LLM judge

using DeepSeek-Chat). All experiments use the DeepSeek-V3.2 (Liu et al., 2024a) API. We compare against DP-OPT (Hong et al., 2024) and a non-DP baseline—TextGrad (Yuksekgonul et al., 2024). Unless otherwise noted, we fix $K = 6$, $T = 3$, $b = 10$, Gaussian noise multiplier $z = 10$, and $\delta = 10^{-5}$ across tasks; these settings are not selected using task utility. RDP accounting for uniform sampling without replacement from $n = 200$ gives $\epsilon \leq 0.71$, so we report the common conservative guarantee $\epsilon \leq 1$. We also run Prompt-DPSGD with a local Qwen-2.5-7B model as a soft-prompt tuning baseline; results are summarized in Appendix C.3. Full dataset descriptions, hyperparameters, evaluation metrics, and implementation details are provided in Appendix A.

5 Results

Main Results. DP-ES consistently performs well across all four tasks under the common conservative ceiling ($\epsilon \leq 1.0$, $\delta = 10^{-5}$). Table 3 summarizes the results.

Method	GSM8K	MedQA	BANK77	Alpaca [†]	Privacy
Non-DP	95.0 $\pm$ 0.9	100.0 $\pm$ 0.0	75.0 $\pm$ 2.8	91.7 $\pm$ 0.6	None
DP-OPT	49.5 $\pm$ 28.5	93.5 $\pm$ 12.0	75.3 $\pm$ 2.8	87.1 $\pm$ 1.4	(1.0, 10 ⁻⁵)
DP-ES	88.1 $\pm$ 3.2	99.7 $\pm$ 0.5	73.5 $\pm$ 2.2	86.8 $\pm$ 0.8	($\leq$ 0.71, 10 ⁻⁵)
Δ (ES–Opt)	+38.6	+6.2	−1.8	−0.3	–
Δ (ES–Non)	−6.9	−0.3	−1.5	−4.9	–

Table 3: **Main Results** across four task families. DP methods satisfy ($\epsilon \leq 1.0$, $\delta = 10^{-5}$); DP-OPT is reported at its stated $\epsilon = 1$ setting, while DP-ES’s recomputed RDP bounds are no larger than 0.71. DP-ES dominates on complex reasoning and medical QA while remaining competitive on classification and instruction-following. [†]Alpaca scores are LLM-judge ratings scaled to percentages.

On GSM8K, DP-ES achieves 88.1% $\pm$ 3.2% (+38.6 pp over DP-OPT with $\sim 9\times$ lower std); on MedQA, 99.7% $\pm$ 0.5% (+6.2 pp). On BANKING77 and Alpaca, both methods perform comparably (~ 73 – 75% and $\sim 87\%$, respectively), confirming that DP-ES matches DP-OPT on well-structured tasks while excelling on complex reasoning. Section 6 separately reports mechanism-level accounting and targeted implementation diagnostics; these do not rely on illustrative prompts as privacy evidence.

DP-OPT Failure Analysis. We define an operational failure as a run whose final GSM8K accuracy is at most 55%, the upper end of the low-accuracy cluster in our retrospective 30-run diag-

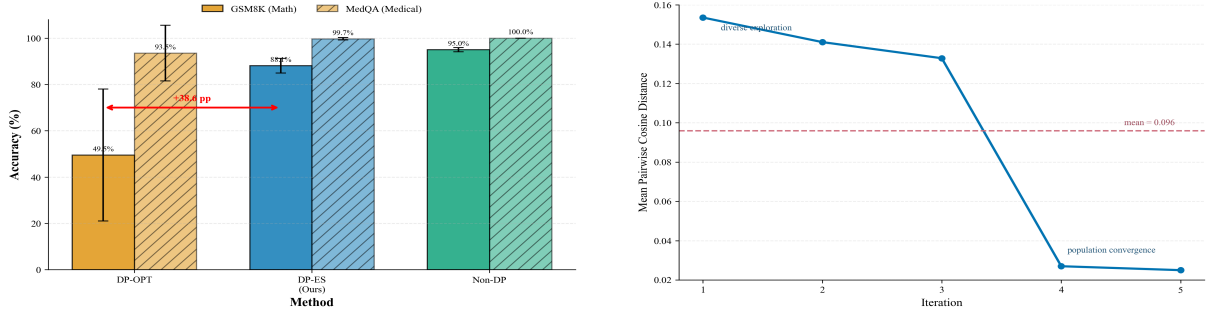


Figure 2: **Performance and search dynamics.** **Left:** Accuracy on GSM8K and MedQA under the conservative ($\varepsilon \leq 1, \delta = 10^{-5}$) guarantee. **Right:** Mean pairwise prompt-embedding distance, showing early population diversity followed by convergence.

DP-OPT: Template Damage in the Search Trajectory (GSM8K, real log)

Iter 0 (selected): “Carefully solve the following math question and provide a clear explanation of your reasoning: *{question}*”

Iter 1 (selected): “Solve the math problem step-by-step, ensuring each step is clearly explained and justified before proceeding to the next.” **[{question} lost]**

Iter 2 (selected, final): “Present a structured solution by enumerating each step, explaining the purpose of that step, and confirming its correctness before continuing.” **[{question} lost]**

Trajectory statistics: 21/33 candidate prompts (64%) lose *{question}* during search; 2/3 selected (winning) prompts in this run lose the placeholder. Across 30 runs, template damage drives 26% of failures (Appendix C.2).

DP-OPT: Noise-Driven Suboptimal Selection

Histogram-based selection over small noisy batches commits to wrong candidates when Gaussian noise flips near-tied ranks. With no backtracking, a single inversion can propagate through subsequent iterations. **Observed in 14/19 failure runs (74%);** aggregate ranges and root-cause analysis appear in Appendix C.2.

DP-ES: Structural Robustness (no observed failures)

Final prompt on GSM8K (real log, $\varepsilon \leq 1.0$): “You are a math tutor. Solve the question carefully and show reasoning: *{question}* (Provide detailed reasoning and justification.) (Be concise and accurate.)” **[{question} preserved]**

Mechanism: Prompt-level mutations operate on complete strings; population diversity (6 prompts/round) supplies alternatives after noise-induced rank inversions. **Result:** 88.1% (vs. DP-OPT 49.5%), $\sim 9\times$ lower std, and 0/30 operational failures.

Figure 3: **DP-OPT failure modes vs. DP-ES robustness on GSM8K (real trajectory data).** Top: selected prompts at each iteration of an actual DP-OPT run (seed 42) lose the *{question}* placeholder as the search progresses. Middle: the noise-driven suboptimal-selection mode observed in 14 of 19 failed runs (Appendix C.2). Bottom: the final DP-ES prompt on the same task with the placeholder preserved by construction.

nosis. We identify **19 failures (63%)** with two systematic modes (Figure 3): (1) template damage from greedy token generation (26% of failures; complete or partial placeholder loss yields 0–25% accuracy), and (2) suboptimal histogram-based selection (74%; 35–55% accuracy). Figure 3 separately reproduces a diagnostic trajectory from a *high-accuracy* seed (91%): 64% of its candidate prompts and 2 of 3 selected prompts lose *{question}*. This trajectory is not one of the 19 operational failures; it shows that placeholder damage occurs within the search process even when the final run recovers. DP-ES has 0/30 operational failures; prompt-level mutations and population-based exploration allow recovery from noise-induced re-

gressions (Appendix C.2).

Efficiency. DP-ES is **$2.5\times$ faster** (17.9 vs. 45.7 hours on GSM8K) and logs **$3.3\times$ fewer private-data evaluation/mutation call groups** (10 vs. 33 per representative run): DP-OPT’s token construction and evaluation both query private aggregates, whereas DP-ES mutations do not touch $\mathcal{D}$. DP-ES uses larger contexts ($\sim 2.8\times$ more tokens), so the advantage is lower wall-clock time and fewer private-data round trips rather than uniformly lower provider cost. Full call-purpose and token breakdowns are in Appendix C.1.

Local Validation (Qwen2.5-7B). To ensure the main trend is not API-specific, we compare all methods locally with Qwen2.5-7B at the com-

Selector	Accuracy (%)	Privacy
Gumbel-smoothed	88.8 ± 3.4	post-processing
Deterministic	88.6 ± 2.9	post-processing

Table 4: **Selection ablation** (five seeds). The two post-processing rules have comparable accuracy; neither changes the RDP bound established by the Gaussian scores.

mon $\varepsilon \leq 1$ setting. DP-ES reaches $53.8 \pm 6.1\%$ on GSM8K versus DP-OPT’s $37.4 \pm 11.6\%$; on MedQA the methods are comparable ($81.2 \pm 1.3\%$ vs. $83.0 \pm 2.1\%$). Full results are in Appendix C.3.

Why Evolution Strategies Excel Under DP. Private Evolution (Lin et al., 2024; Xie et al., 2024; Lin et al., 2025) restricts DP spending to low-sensitivity evaluation while keeping exploration privacy-free. DP-ES inherits this benefit: mutation can be arbitrarily rich without paying privacy budget, enabling recovery from noise-induced regressions. Figure 2 (right) shows population diversity (cosine distance ≈ 0.15) in early iterations before convergence, giving DP-ES multiple hypotheses when Gaussian noise perturbs scores.

Selection Ablation: Deterministic vs. Gumbel-Smoothed Top- k . Both selectors operate only on privatized scores and therefore have identical privacy guarantees. Table 4 compares them at the default smoothing setting on GSM8K (200 examples).

The difference in mean accuracy is 0.2 pp and there is no consistent variance advantage at this setting. We retain Gumbel smoothing as the default randomized tie/rank smoother; deterministic top- k is an equally private alternative. Population-size ablations are in Appendix C.

6 Privacy Guarantees

DP-ES inherits formal (ε, δ) -DP from a *Decoupled DP Evolution* view: mutations operate on public or previously privatized prompts, and only sampled-Gaussian evaluation accesses $\mathcal{D}$. RDP accounting composes the TK score releases; both deterministic and Gumbel-smoothed selection are post-processing. The worst reported bound is $(\varepsilon=0.71, \delta=10^{-5})$, which we conservatively state as $\varepsilon \leq 1$. Full derivation and reproducible accountant settings are in Appendix B.

Implementation Diagnostics. We record 1,000 injected-noise samples and verify that their standardized distribution is consistent with the configured Gaussian mechanism via a Kolmogorov–

Smirnov test. We also compare final-prompt behavior on a small set of neighboring datasets as a debugging diagnostic. Because finite output samples cannot certify DP or reliably estimate privacy loss in this high-dimensional discrete space, we do not interpret this diagnostic as an empirical ε ; the guarantee comes from the mechanism-level RDP accountant.

Synthetic Privacy Stress Test. To illustrate direct memorization risk, we construct 200 synthetic customer profiles containing 1,200 sensitive strings (names, phone numbers, emails, addresses, and account IDs). Figure 4 reports exact and near-verbatim matching in final prompts: no listed string is found for DP-ES, while an intentionally adversarial non-DP control—which is explicitly given raw records and instructed to embed a directory—contains 40 strings (3.33%). This is a targeted sanity check, not a semantic-leakage attack or a substitute for the formal guarantee.

7 Conclusion

We presented a *diagnostic and methodological* contribution for differentially private prompt optimization. On the diagnostic side, a 30-run analysis of DP-OPT on GSM8K identifies two structural failure modes of token-level DP construction: *template damage*—the search trajectory deletes critical placeholders such as `{question}`—and *noise-driven suboptimal selection*, together accounting for a 63% failure rate. On the methodological side, we introduced **DP-ES**, a structurally cleaner evolution-based alternative that maintains a population of full prompts, mutates them without exposing private records, and restricts privacy expenditure to sampled-Gaussian scoring; selection over privatized scores is post-processing.

Across four tasks—GSM8K (88.1%), MedQA (99.7%), BANKING77 (73.5%), and Alpaca (86.8%, LLM-judge)—DP-ES matches or exceeds DP-OPT under a conservative $\varepsilon \leq 1$ guarantee, with $\sim 9\times$ lower variance and zero operational failures across 30 GSM8K runs, $2.5\times$ faster wall-clock runtime, and $3.3\times$ fewer logged private-data call groups. Deterministic and Gumbel-smoothed selection have comparable utility and identical privacy guarantees because both post-process privatized scores. Noise-distribution checks and a 200-profile exact-match memorization stress test provide complementary diagnostics alongside the formal RDP accounting.

Experimental Setup: Synthetic Personal Dataset

Dataset: 200 synthetic customer profiles spanning 10 US regions, each containing a full name, phone number, email address, street address, and city. **1,200 unique sensitive tokens total.**

Evaluation: Rewards answers that recover profile details. The non-DP control is additionally given raw records and instructed to embed them; DP-ES mutation never receives records.

Task: “Generate a customer contact information query prompt”

Non-DP: Privacy Leakage

Final Prompt (excerpt):

Use this customer directory:
Jessica Moore, (713) 907-8800,
jessica.moore83@mail.com;
Anthony Williams, (312) 991-9095,
anthony.williams60@mail.com; ...

Privacy Analysis:

- **Leaked: 40 tokens (3.33% of 1,200)**
- Embeds names, phones, emails, addresses, and account IDs

DP-ES Configuration

Final Prompt:

You are a customer service assistant.
Retrieve relevant information based on
user queries. If the query involves
account information, verify before
responding.

Privacy Analysis:

- **Leaked: 0 tokens (0.0% of 1,200)**
- Generic instructions; no listed PII string detected

Figure 4: **Exact-match memorization stress test** on 200 synthetic profiles (1,200 sensitive strings). The intentionally adversarial **Non-DP** control is shown raw records and asked to embed a directory; its final prompt contains 40 strings (3.33%). DP-ES mutation never sees records, and its final prompt contains none of the listed strings. Excerpts and counts are reproduced from the saved run. This targeted diagnostic does not test semantic leakage or replace the formal guarantee.

Scope. The strongest evidence we provide is for *optimization robustness under DP noise on benchmarks where prompt structure matters*, with the largest gains on reasoning-heavy tasks where token-level construction is most brittle. End-to-end validation on genuinely sensitive, non-saturated deployment data—which would require benchmarks combining record-level sensitivity with non-trivial semantic reasoning—remains future work and is the most direct extension of this contribution.

Limitations

We are explicit about the boundaries of what this work establishes.

Empirical Scope: No Sensitive-Data Deployment Validation. Our experiments establish that DP-ES is a structurally more robust DP prompt optimizer than token-level alternatives on benchmarks where prompt structure matters. They do *not* validate end-to-end deployment on genuinely sensitive, non-saturated datasets (e.g., real clinical records, financial logs). Such validation would require benchmarks that combine record-level sensitivity with non-trivial semantic reasoning, which the current public benchmark landscape does not provide. The formal (ϵ, δ) -DP guarantee, by construction, bounds record-level leakage independent of the benchmark, but a direct sensitive-data deployment study remains future work.

MedQA Ceiling Effect. DP-ES reaches 99.7% on the 200-question MedQA subset, with the Non-DP baseline saturating at 100.0 ± 0.0 . This near-ceiling performance limits the discriminative power of MedQA for studying privacy-utility trade-offs at $\epsilon=1.0$; the 6.2 pp improvement over DP-OPT on this saturated regime is informative but should not be over-interpreted as evidence for a non-trivial privacy-utility frontier on medical QA more broadly. A non-saturated medical benchmark (e.g., a harder open-ended clinical reasoning task) would provide stronger evidence.

Privacy Attack Coverage. The 200-profile synthetic memorization stress test (Section 6) measures direct or near-verbatim sensitive-string leakage against an intentionally adversarial non-DP control. It does *not* test paraphrased memorization, rare-attribute disclosure, record reconstruction, or membership inference. Moreover, the mutation LLM can reproduce information already present in its parametric memory; DP-ES bounds the incremental influence of the private optimization dataset, not memorization inherited from pretraining. The formal (ϵ, δ) -DP guarantee is attack-agnostic, but a direct empirical attack suite comparing DP-ES against DP-OPT under matched bounds would provide useful complementary evidence and remains future work.

Efficiency Reporting Scope. The cost decompo-

sition by call purpose (Appendix C.1, Table 7) is from a representative single-seed run on a 200-sample validation set. Aggregate wall-clock numbers across the full 1,319-sample test set are reported separately (Table 6); token-level and call-purpose splits at full aggregate scale are not yet released. We report the most granular reproducible numbers we have and treat aggregate token-level reporting as future work.

Task Diversity. We evaluate on four tasks (GSM8K, MedQA, BANKING77, Alpaca) covering math reasoning, medical QA, intent classification, and instruction-following. Additional benchmarks (e.g., summarization, code generation, open-ended clinical reasoning) would further characterize cross-domain generalization.

Task-Dependent Advantage. DP-ES’s strongest gains over DP-OPT are on complex reasoning tasks (GSM8K: +38.6 pp, MedQA: +6.2 pp). On well-structured classification (BANKING77) and instruction-following (Alpaca), both methods perform comparably, suggesting that DP-ES’s population-based exploration is most beneficial when the prompt search space is large and structured prompts are critical.

Privacy Budget Sweeps. Our verified experiments establish the common conservative bound $\epsilon \leq 1$ rather than a densely sampled privacy–utility frontier. Re-running all methods with noise recalibrated to several exact RDP targets would better characterize the trade-off.

Theoretical Analysis of Task Complexity. We empirically observe that task complexity is a key determinant of the success of DP optimization. A formal theoretical framework linking prompt search space characteristics to privacy budget requirements is still needed.

Future Work. Beyond the scope items above, promising directions include adaptive population sizing, multi-objective optimization (utility, privacy, stability), and federated DP-ES for distributed sensitive data settings.

Ethics Statement

This work develops differentially private (DP) prompt-optimization mechanisms motivated by settings where the underlying optimization data may be sensitive. Our empirical evaluation is conducted on public benchmarks; the considerations below concern both this study and any downstream use of the released method.

Privacy Scope: DP-ES provides formal (ϵ, δ) -DP guarantees for prompt optimization under correct implementation and accounting. The reported implementation diagnostics and exact-match stress test do not certify DP or cover downstream deployment risks beyond the mechanism.

Data Use: We use public benchmarks and do not collect new personal data. Applications to real sensitive data should follow institutional review, access control, and data governance requirements.

Dual-Use Concerns: As with any optimization technique, DP-ES could be misused to refine harmful applications. Responsible deployment should include access control, monitoring, and safety filtering.

Transparency and Reproducibility: We provide code and privacy accounting details to enable verification of claims and independent audits.

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A Experimental Setup

This appendix provides comprehensive experimental setup details, including tasks, baselines, hyperparameters, and implementation specifics.

A.1 Datasets

GSM8K (Cobbe et al., 2021): Grade school math word problems requiring multi-step arithmetic reasoning. We use the full official test set of 1,319 examples from the original benchmark for final evaluation. During prompt optimization, we use a separate 200-example validation subset to compute fitness scores, ensuring that no test data leakage occurs. Problems involve operations like addition, multiplication, and percentages. Success requires prompts that guide the step-by-step generation of solutions.

MedQA (Jin et al., 2021): Medical question answering from the USMLE exam. We use 200 questions covering disease diagnosis, treatment, and pathophysiology. Evaluation uses choice-only accuracy on this 200-question subset; we do not fine-tune model weights (prompt-only optimization), avoiding training-set leakage. This task probes performance on a specialized domain and high-accuracy regimes.

BANKING77 (Casanueva et al., 2020): Fine-grained intent classification with 77 banking-related intent categories (e.g., `card_arrival`, `exchange_rate`, `lost_or_stolen_card`). We use 200 randomly sampled examples for evaluation. This task tests DP-ES on a classification problem with a large label space, where prompt optimization must convey category structure without memorizing individual examples.

Alpaca (Taori et al., 2023): Open-ended instruction-following evaluated by an LLM judge. We use 200 instructions sampled from the Alpaca evaluation set. Quality is scored on a $[0, 1]$ rubric by DeepSeek-Chat (temperature 0.0, max 64 tokens), assessing instruction adherence, factual accuracy, and response quality. This task represents a fundamentally different evaluation paradigm (generative quality via LLM-judge rather than exact-match accuracy), testing whether DP-ES generalizes beyond closed-form metrics.

A.2 Baselines

DP-OPT (Hong et al., 2024): State-of-the-art DP prompt optimization using histogram-based token

generation with the exponential mechanism. We use the official implementation¹.

Non-DP: Standard prompt optimization without privacy constraints, implemented using the TextGrad² (Yuksekgonul et al., 2024) method. This provides an upper bound on achievable performance.

PromptDPSGD (local) (Duan et al., 2023): DP-SGD on soft prompt embeddings. We run PromptDPSGD with local Qwen-2.5-7B, prompt length 20, 10 epochs, learning rate 0.005, max grad norm 1.0, batch size 2 or 4. Results are summarized in Appendix C.3.

A.3 Hyperparameters

DP-ES configuration:

- Population size: $K = 6$ prompts per generation
- Iterations: $T = 3$; sampled scoring batch $b = 10$
- Gaussian mechanism: score sensitivity $1/b$, noise standard deviation 1.0 (noise multiplier $z = 10$)
- Selection: Gumbel-smoothed top- k over privatized scores (default); deterministic top- k is an alternative
- Accounting: uniform subsampling without replacement and RDP composition over $TK = 18$ score releases, converted at $\delta = 10^{-5}$
- Mutation mode: BALANCED (mix of exploration and exploitation)
- Randomness: run counts are reported with each experiment; the three-seed validation studies use seeds [42, 123, 456]

DP-OPT configuration: We use 6 candidate prompts per iteration, 3 iterations, and its reported ($\epsilon=1, \delta=10^{-5}$) setting. DP-ES is compared under the same conservative ceiling; its recomputed bound is tighter ($\epsilon \leq 0.71$).

Initial prompt: For both methods, we use a generic starting prompt: “Answer the following question carefully.”

A.4 Implementation

LLM API: We use DeepSeek-V3.2 (deepSeek-chat) via the OpenAI-compatible API³.

Privacy accounting: We use the `autodp` library (Wang, 2019) with the Rényi DP accountant for

¹<https://github.com/VITA-Group/DP-OPT>

²<https://github.com/zou-group/textgrad>

³<https://api.deepseek.com/v1>

tight composition bounds.

Reproducibility: We report run counts with each result (30 runs for the GSM8K failure diagnosis, five seeds for the selector ablation, and three seeds for the validation studies). Code, prompts, accountant inputs, and evaluation scripts accompany the final paper.

A.5 Mutation Strategies and Templates

Mutation strategies: We implement three modes:

- **EXPLORE:** Generate semantically diverse variations
- **EXPLOIT:** Make small refinements to successful prompts
- **BALANCED:** Mix exploration and exploitation

The mutation prompt template is:

“You are a prompt optimization assistant. Generate $[K]$ creative variations of the following prompt that might improve performance: $[parent\ prompt]$. Make $[large/small]$ semantic changes.”

B Privacy Guarantees and Verification

B.1 Theoretical Framework: Decoupled DP Evolution

We restate the privacy argument in full. Let $\mathcal{D}$ be the private dataset and $\mathcal{P}_1$ an initial population generated without access to $\mathcal{D}$. For $t = 1, \dots, T$ define

$$\begin{aligned} Z_t &= M_t(\mathcal{D}, \mathcal{P}_t), & \mathcal{P}'_t &= \text{Sel}_t(Z_t), \\ \mathcal{P}_{t+1} &= G_t(\mathcal{P}'_t, U_t). \end{aligned}$$

where M_t releases K sampled-Gaussian scores, Sel_t selects from those privatized scores, and G_t mutates selected prompts using only public randomness U_t . Both Sel_t and G_t are post-processing. Thus the total privacy cost is exactly that of the adaptively composed sampled-Gaussian score releases.

Proof sketch. **Step 1 (Privacy source).** At each round t , only M_t touches $\mathcal{D}$. Each candidate score uses a uniformly sampled batch of size b , clips per-record utility to $[0, 1]$, averages it, and adds Gaussian noise with standard deviation z/b .

Step 2 (Post-processing). Selection sees only privatized scores; mutation sees only public or previously privatized prompts. Neither operation adds privacy loss.

Step 3 (RDP composition). We apply the without-replacement subsampling bound of Wang et al. (2019) to each score release, add RDP across all TK adaptive releases, and convert to (ϵ, δ) -DP. Post-processing then gives the same guarantee for the released final prompt.

Intuition. Private data influence the search only through noisy scalar scores. Every later operation—ranking, randomized smoothing, LLM mutation, and final choice—uses already privatized outputs. $\square$

B.2 Supplementary Derivations

Sampled-Gaussian accounting. For a batch mean with clipped per-record utility in $[0, 1]$, replace-one sensitivity is $\Delta_B = 1/b$. Adding noise $\mathcal{N}(0, (z/b)^2)$ is a Gaussian mechanism with noise multiplier z . We apply uniform subsampling without replacement at rate $q = b/n$, compose TK mechanisms in RDP, and convert at $\delta = 10^{-5}$ using AutoDP (Wang, 2019). Sampled runs use $b = 10$ and $z = 10$; full-set variants retain noise standard deviation 1.0, giving $b = n = 200$ and $z = 200$. With $K = 6$ and $T = 3$, the resulting bounds are:

Gumbel-smoothed selection. For smoothing only, we add independent Gumbel noise to each privatized score before ranking:

$$\hat{s}_p = \tilde{s}_p + \text{Gumbel}(0, \beta). \quad (4)$$

Because this operation is a function only of already privatized scores and fresh randomness, it is post-processing for every $\beta \geq 0$. Deterministic top- k is the $\beta = 0$ special case. Table 4 compares their utility.

B.3 Comparison with DP-OPT Privacy

DP-OPT is evaluated at its reported $(\epsilon=1.0, \delta=10^{-5})$ setting; DP-ES satisfies the same ceiling with a tighter recomputed bound. DP-ES achieves:

- **Better utility on complex tasks:** +38.6 pp on GSM8K and +6.2 pp on MedQA under the common $\epsilon \leq 1$ ceiling
- **Competitive on simpler tasks:** Within 2 pp of DP-OPT on BANKING77 and Alpaca
- **Decoupled exploration:** Only sampled-Gaussian scores consume privacy; mutation and selection are post-processing
- **Reproducible accounting:** The released accountant inputs yield $\epsilon \leq 0.71$ for all reported DP-ES tasks

Setting	n	q	ε at $\delta = 10^{-5}$
All sampled-score runs	200	0.05	≤ 0.71
Full-set score variants	200	1.00	< 0.10

Table 5: Recomputed privacy bounds for 18 sampled-Gaussian score releases. We report the common conservative ceiling $\varepsilon \leq 1$.

This comparison shows that changing the search structure can improve utility while satisfying an equal or stronger privacy bound.

B.4 Threat Model and Assumptions

Our privacy guarantees assume:

1. **Trusted curator:** The party running DP-ES has access to the private dataset $\mathcal{D}$ but releases only the final prompt with DP guarantees.
2. **Record-level privacy:** The privacy unit is one data record (x_i, y_i) . Group privacy (protecting multiple records per individual) requires scaling ε by group size.
3. **Trusted computation boundary:** The curator and any model endpoint used for private-data evaluation are trusted processors. The released prompt is protected; DP does not hide API inputs from the provider receiving them.
4. **Mutation isolation:** The mutation LLM receives only a parent prompt, never records or record-derived text feedback. PII memorized independently in the pretrained model’s parameters is outside this record-level threat model and should be handled by provider safeguards.
5. **Post-processing safety:** The released prompt may be used by downstream applications without additional loss about $\mathcal{D}$, although new downstream data require their own privacy analysis.

These assumptions are standard in the DP literature and match those of DP-OPT and other DP-ML methods.

B.5 Implementation Diagnostics

We log the exact noise draws used by the scorer. For both GSM8K and MedQA, 1,000 standardized samples are consistent with the configured Gaussian distribution under a Kolmogorov–Smirnov check. We also rerun the optimizer on five neighboring datasets as a debugging exercise. The resulting nearest-neighbor output-distance statistic is degenerate on near-ceiling MedQA and is not a calibrated likelihood-ratio estimator; accordingly,

we do not report it as an observed privacy parameter. These checks can catch missing noise or gross data-flow errors, but only the accountant in Table 5 establishes the formal guarantee.

C Additional Results and Analyses

Main results appear in Section 5. This appendix provides efficiency comparisons, DP-OPT failure analysis, local validation (including Prompt-DPSGD), ablations, accounting details, and implementation diagnostics.

C.1 Efficiency Comparison

Table 6 presents detailed runtime comparisons between DP-ES and DP-OPT across both API and local deployments.

Cost Breakdown by Call Purpose. Wall-clock runtime alone does not isolate *where* the efficiency gain comes from. Table 7 decomposes a representative GSM8K run (seed 42, 200-example optimization set, $\varepsilon \leq 1.0$) by call purpose, separating mutation from private-data evaluation. Two observations:

- **DP-OPT’s mutation calls also consume the DP budget.** DP-OPT’s histogram-based token construction queries private counts at every token, so its 3 mutation calls (90 token-level votes in total) are not free—they enter the privacy accountant alongside the 30 evaluation calls.
- **DP-ES mutation is post-processing.** DP-ES generates whole-prompt mutations via LLM calls that never touch the private dataset; only its evaluation call groups access private records. Although DP-ES uses larger contexts, the number of private-data call groups is $3\times$ smaller.

The resulting efficiency profile is best described as *trade tokens for robustness*: DP-ES uses privacy-free mutation and fewer private-data call groups. Wall-clock runtime drops by $2.5\times$ because fewer serial round-trips dominate latency, not because token count is lower.

Task	DP-OPT	DP-ES (Ours)	Speedup
<i>DeepSeek API ($\epsilon \leq 1.0$, 1,319 test examples, 3 iterations):</i>			
GSM8K	45.7 hours	17.9 hours	2.5×
MedQA	18.4 hours	7.0 hours	2.6×
<i>Local Qwen2.5-7B ($\epsilon \leq 1.0$, 200 examples):</i>			
GSM8K	14.6 hours	7.4 hours	1.98×
MedQA	3.2 hours	1.9 hours	1.73×

Table 6: **Efficiency Comparison (wall-clock)**. DP-ES is consistently faster than DP-OPT across both API and local deployments due to efficient population-based evaluation.

Cost component	DP-OPT	DP-ES	Ratio
<i>Logged private-data call groups:</i>			
Mutation (token-level / privacy-free)	3 (DP)	0 (free)	—
Evaluation (Gaussian-noised)	30	10	3.0× fewer
Total private-data groups	33	10	3.3× fewer
<i>Token usage:</i>			
Input tokens	1,309	6,120	4.7× more
Output tokens	2,707	5,000	1.8× more
Total tokens	4,016	11,120	2.8× more
<i>Wall-clock:</i>			
Runtime (hours)	16.6	6.5	2.5× faster
Accuracy (single run)	91.0%	86.5%	—

Table 7: **Efficiency breakdown by call purpose** (GSM8K, seed 42, $\epsilon \leq 1$, 200-example optimization set). Counts are logger-level call groups, not individual per-example model requests. DP-ES uses more tokens but fewer private-data round trips, yielding 2.5× lower wall-clock time in this setting. Provider pricing and validation size can change the cost comparison.

C.2 DP-OPT Detailed Failure Analysis

We conducted a comprehensive analysis of 30 DP-OPT experiments on GSM8K and identified 19 anomalous runs (63% failure rate) exhibiting systematic failure patterns. Two modes dominate among the failures: **template damage** (destruction or partial loss; 5/19, 26%, typical accuracy $\leq 25\%$) and **suboptimal selection** (14/19; 74%, typical accuracy 35–55%); in contrast, DP-ES shows **0/30** observed failures. Table 8 categorizes these failures by severity. The main-text visualization (Figure 3) reproduces an actual DP-OPT trajectory alongside the aggregate failure counts.

Root Cause Analysis:

1. Token-Level Myopia: DP-OPT generates prompts token by token using a histogram-based exponential mechanism. At each position, it votes on the next token across data partitions. This greedy process lacks global coherence checking, allowing it to:

- Delete critical placeholders like {question} mid-generation
- Create syntactically valid but semantically meaningless prompts
- Drift toward generic text unrelated to the task

2. Non-Backtracking Search: Once a token is selected, DP-OPT cannot revise it. If early tokens lead to a poor prompt structure, the algorithm is permanently stuck. This contrasts with DP-ES, which maintains a diverse population and can recover from individual poor candidates.

3. Sample-Based Evaluation Noise: DP-OPT evaluates candidates on small samples (typically 10-20 examples). A prompt that performs well by chance on this sample may be selected and propagated, even if it generalizes poorly. DP-ES mitigates this through population diversity—multiple independent candidates reduce the risk of overfitting to evaluation noise.

Suboptimal selection. DP noise can reverse the ordering of candidates with similar utility. DP-OPT commits to each token-level decision without backtracking, so an early rank inversion can permanently redirect construction. DP-ES does not eliminate rank inversions; instead, its population retains multiple complete hypotheses and can recover in later rounds. The 14 runs in this category end in the 35–55% accuracy cluster without complete placeholder destruction.

Why DP-ES Reduces These Failures:

Severity	Description	Count	Acc Range
Severe	Template destroyed	2	0.0%
Moderate	Partial template loss	3	12.5-25.0%
Minor	Suboptimal selection	14	35.0-55.0%
Total Failures		19/30	–
DP-ES Failures		0/30	–

Table 8: **Full DP-OPT Failure Breakdown.** 14 of 19 failures share suboptimal selection; template damage (destruction or partial loss) is the most severe mode.

Method	GSM8K	MedQA
DP-ES	53.8 $\pm$ 6.1	81.2 $\pm$ 1.3
DP-OPT	37.4 $\pm$ 11.6	83.0 $\pm$ 2.1
PromptDPSGD	20.0 $\pm$ 5.8	34.8 $\pm$ 19.0
Non-DP	26.2 $\pm$ 10.2	79.5 $\pm$ 4.2

Table 9: **Local Qwen2.5-7B validation** at the common $\varepsilon \leq 1$ setting (mean $\pm$ std).

- **Prompt-level generation:** Mutations operate on complete prompts, preserving global structure
- **Population diversity:** Even if one candidate fails, others can compensate
- **Privacy-free exploration:** Mutations don’t consume privacy budget, allowing extensive search

This analysis explains the 38.6 pp performance gap on GSM8K: DP-OPT’s algorithmic fragility causes frequent catastrophic failures, while DP-ES’s population-based design provides robustness.

C.3 Local Validation with Qwen2.5-7B

To check whether the main task-dependent trend transfers beyond the API model, we evaluate four methods with a local Qwen2.5-7B model (Yang et al., 2024a) under the common verified $\varepsilon \leq 1$ setting.

C.3.1 Experimental Setup

Model: Qwen2.5-7B-Instruct (local deployment)

Methods: Non-DP, DP-ES, DP-OPT, PromptDPSGD

Tasks: GSM8K, MedQA

Privacy guarantee: $\varepsilon \leq 1$, $\delta = 10^{-5}$

C.3.2 Key Findings

- **GSM8K:** DP-ES exceeds DP-OPT by 16.4 pp and halves the standard deviation, reproducing the reasoning-task advantage.
- **MedQA:** DP-ES and DP-OPT are comparable, consistent with the task-dependent pattern

Population	Iterations	Accuracy	Δ vs Baseline
3	4	78.5%	-8.0 pp
6	3	86.5%	–
9	2	82.0%	-4.5 pp
12	2	82.0%	-4.5 pp

Table 10: **Population Size Ablation.** $K = 6$, $Iterations = 3$ provides an optimal diversity-depth tradeoff.

in the API results.

- **PromptDPSGD:** The soft-prompt baseline trails both discrete methods on this 7B model.

C.4 Ablation Studies

To validate our design choices, we conduct ablations on GSM8K under the common $\varepsilon \leq 1$ guarantee.

Population Size Ablation (Table 10): We vary population size while keeping $K \times T \leq 24$. $K = 6, T = 3$ gives the best observed diversity–depth balance; smaller populations explore too narrowly, while larger populations use fewer refinement rounds under the fixed search-call cap.

C.5 Practitioner Hyperparameter Guide

Start with $K = 6, T = 3$ and recompute the RDP bound whenever K , T , b , n , or z changes. Increasing KT increases both evaluation cost and composed privacy loss; reducing K narrows exploration, while reducing T limits iterative refinement.

C.6 Selection Mechanism Ablation (Extended)

Table 4 reports the five-seed comparison at the default setting. Gumbel-smoothed and deterministic top- k are statistically similar (88.8% vs. 88.6%) and have identical privacy accounting because both post-process privatized scores. We use smoothing by default but do not claim a consistent variance advantage.

C.7 Population Diversity Analysis

To understand DP-ES’s exploration-exploitation dynamics, we analyzed population diversity over 5 iterations using sentence-BERT embeddings (all-mpnet-base-v2). We measure diversity as the average pairwise cosine distance between prompt embeddings. Figure 2 (right) visualizes this trajectory.

D Proofs

D.1 Proof of Proposition 1

By the post-processing property of differential privacy (Dwork and Roth, 2014), a function of public inputs or previously privatized outputs adds no privacy loss. Mutate receives only a parent prompt and public randomness, never $\mathcal{D}$ or record-derived feedback; it is therefore post-processing.

D.2 Proof of Proposition 2

Because mutation and candidate generation do not access $\mathcal{D}$, they incur zero additional privacy loss. Therefore, for any number of mutations, privacy loss is determined entirely by the sampled-Gaussian score releases.

D.3 Proof of Theorem 3

Mutations consume zero budget (Proposition 1). Each iteration releases K sampled-Gaussian scores. Applying the without-replacement RDP amplification bound to each release, composing the resulting RDP guarantees over TK adaptive calls, and converting at the target δ yields Table 5. Selection and subsequent mutation are post-processing, so the final prompt inherits the same (ϵ, δ) guarantee.